Andreas Tzitzikas

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Education

University of Maryland, College Park, MD — Expected May 2027

Master of Science in Computer Engineering — **GPA: 4.00**

Bachelor of Science in Computer Engineering (GPA: 3.88) — Expected May 2026

Relevant Coursework: Parallel Computing (CMSC416), Algorithms (CMSC351), Organization of Programming Languages (CMSC330), Object-Oriented Programming (CMSC132), Computer Systems (CMSC216), Discrete Structures (CMSC250)

Projects

CPU Performance Analysis Framework | Python, Bash, Automation — Aug 2025 - Present

- Developed comprehensive benchmarking suite in Python for automated performance analysis, implementing statistical analysis and workload characterization across different CPU architectures.
- Architected modular analysis framework with custom scripts for IPC measurement, performance profiling, and automated report generation with detailed trade-off analysis.
- Built scalable testing infrastructure supporting parallel execution and data visualization for comparative architecture studies and optimization research.

Parallel Game of Life Implementation | MPI, C/C++, Performance Optimization — Jan 2025 – May 2025

- Implemented high-performance parallel version of Conway's Game of Life using MPI with 1D row decomposition, achieving scalable performance across 128 processes with 2x reference throughput.
- Architected non-blocking Isend/Irecv communication patterns with ghost cell synchronization and dynamic load balancing for optimal parallel efficiency.
- Developed comprehensive timing analysis framework using MPI_Wtime for performance characterization and bottleneck identification in distributed computing environments.

Space Invaders Video Game | Java, Object-Oriented Design, AWT Graphics — Spring 2025

- Designed complete Space Invaders game using advanced OOP principles including inheritance, polymorphism, inner classes, and interfaces with lambda expressions for bonus systems.
- Implemented real-time graphics engine with collision detection algorithms, coordinated enemy movement patterns, and progressive difficulty scaling using prime number-based firing rates.
- Architected modular game architecture with 15 enemies, boss battles, scoring mechanics, and complete win/lose conditions demonstrating comprehensive software design patterns.

STM32 Bare-Metal Peripheral Integration | ARM Assembly, Embedded C, Real-Time Systems — Jan 2025 – May 2025

- Developed low-level firmware in ARM Assembly and C for STM32G4 microcontroller, implementing register-level programming for GPIO, ADC, DAC, PWM timers, and SPI flash memory interfaces.
- Architected multiplexed interrupt system routing single hardware interrupt to four distinct peripheral drivers using state-based arbitration for efficient resource management.
- Engineered real-time embedded software with non-blocking drivers, precise timing control, and robust error handling for industrial control applications requiring deterministic behavior.

Experience

Embedded Systems Intern, Alchemity – College Park, MD — Jun 2025 – Aug 2025

- Developed full-stack automation suite using Python and Rust to parallelize testing workflows, converting multi-day manual experiments into overnight automated runs with significant throughput improvements.
- Built end-to-end control interfaces including desktop GUIs, CLI tools, and embedded communication protocols to streamline workflows between hardware and software teams.
- Implemented real-time data processing and visualization systems with performance monitoring, error handling, and user-friendly interfaces for experimental control.

Skills

Programming Languages: Python, C/C++, Java, Rust, OCaml, Bash, MPI, OpenMP, CUDA
Software Development: Object-Oriented Design, Algorithms, Data Structures, Parallel Computing, Performance Analysis, Git, Linux
Frameworks & Tools: MPI/OpenMP/CUDA, JUnit Testing, Build Automation, Version Control, Debugging, Profiling
Methodologies: Agile Development, Test-Driven Development, Code Review, Documentation, Algorithm Analysis